Trainning Program in Neuroscience

From neuroimaging to the human connectome

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Valparaíso, August 18, 2026

About me

- Postdoctoral Researcher, Universidad San Sebastián, Chile
- Ph.D. in Engineering Sciences (Electrical Engineering), Universidad de Concepción, Chile
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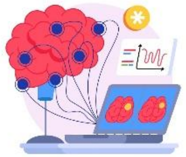


Research Line

Computational Neuroscience

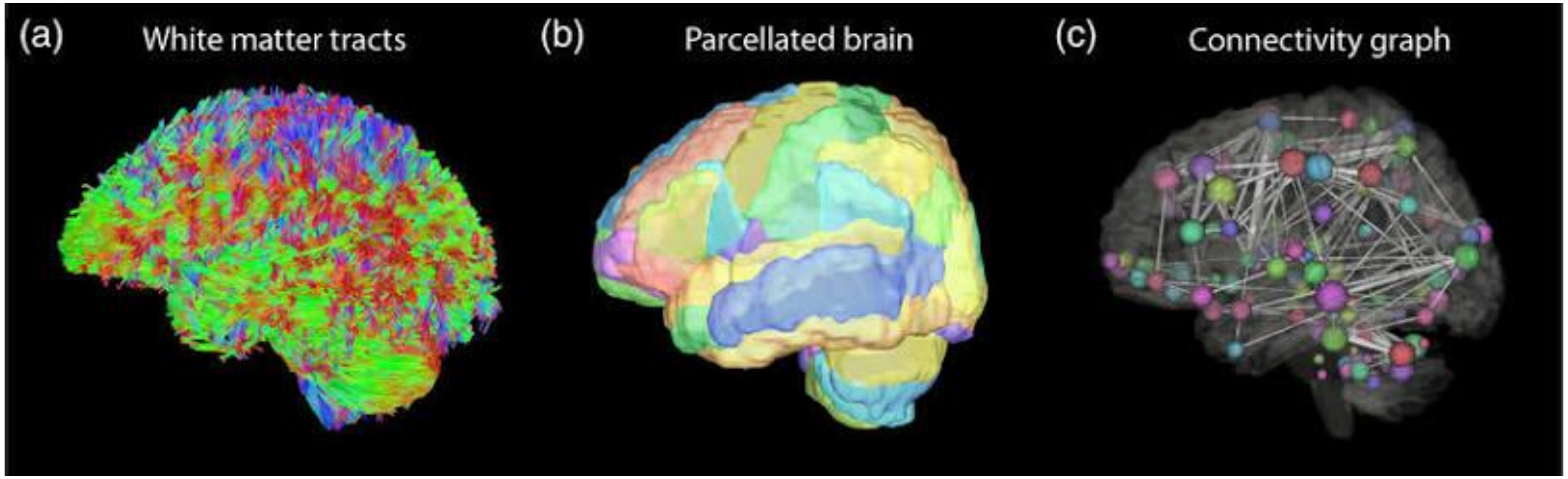
- Brain Connectivity
- Neuroimaging
- Computational Modeling
- Machine learning

CompNeuroUSS

What Is the Connectome?

“The human connectome is a map of the brain’s structural connections, represented as a network of interconnected brain regions.”

Based on Sporns et al., 2005; Sporns, et al., 2013



Reginold et al, 2019

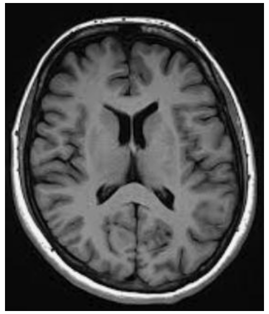
How do we obtain the information needed to build this map?

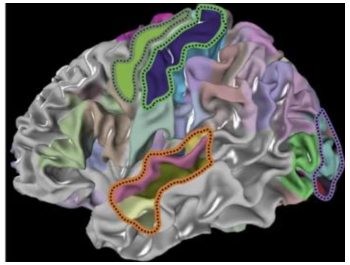
How Do We Observe the Brain?

Magnetic Resonance Imaging (MRI)

Structural MRI

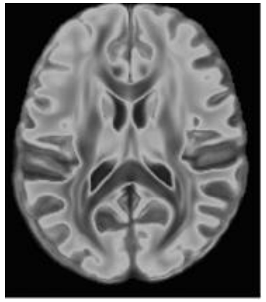
sMRI
(T1-weighted)

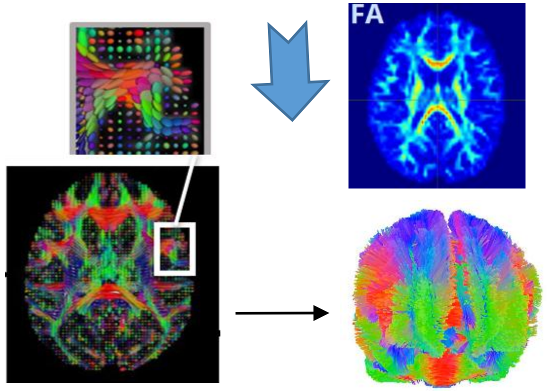




Brain anatomy

Diffusion MRI(dMRI)

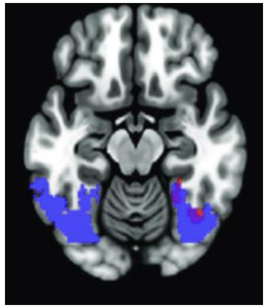


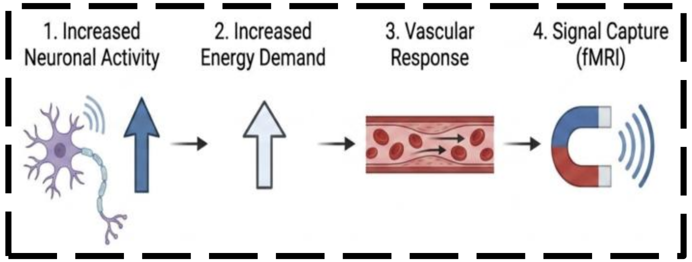


Structural connectivity

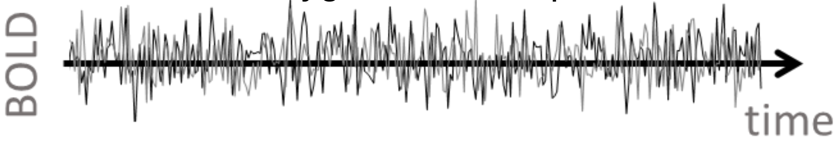
Functional MRI

Resting state fMRI



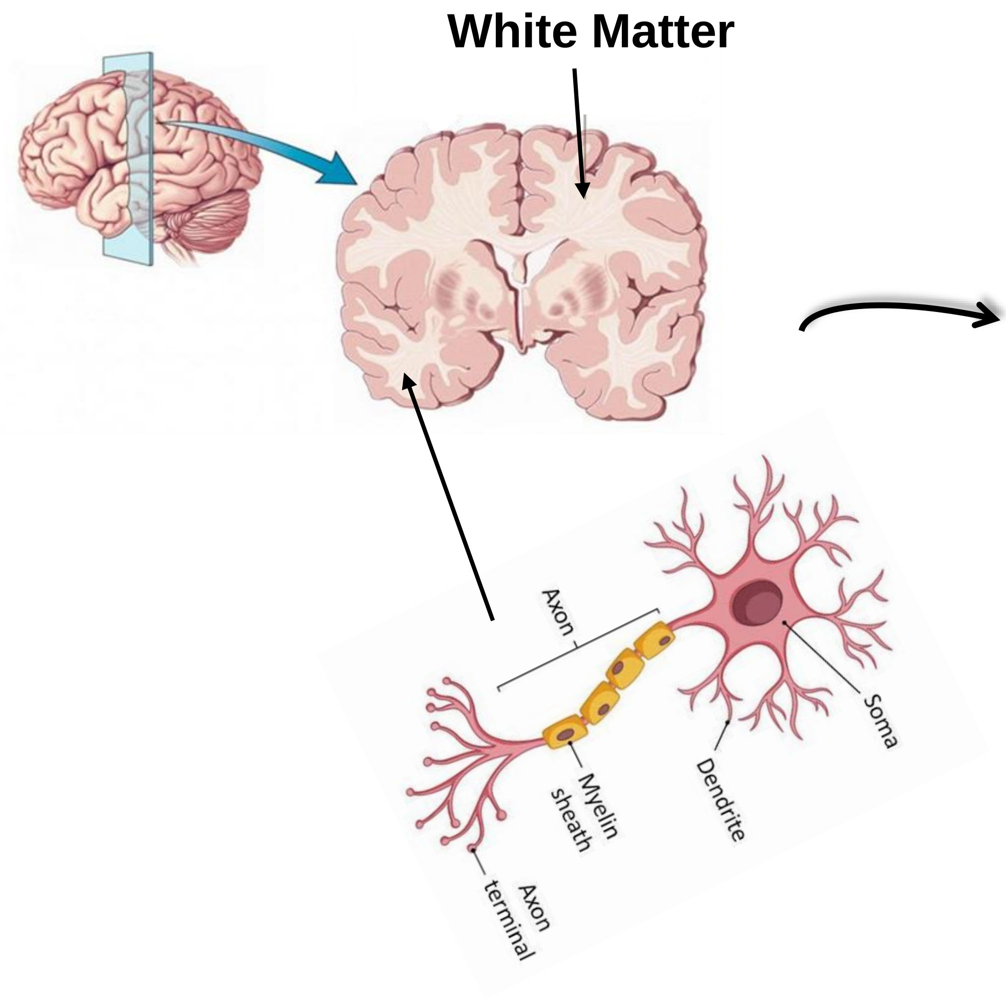


Blood Oxygen Level–Dependent

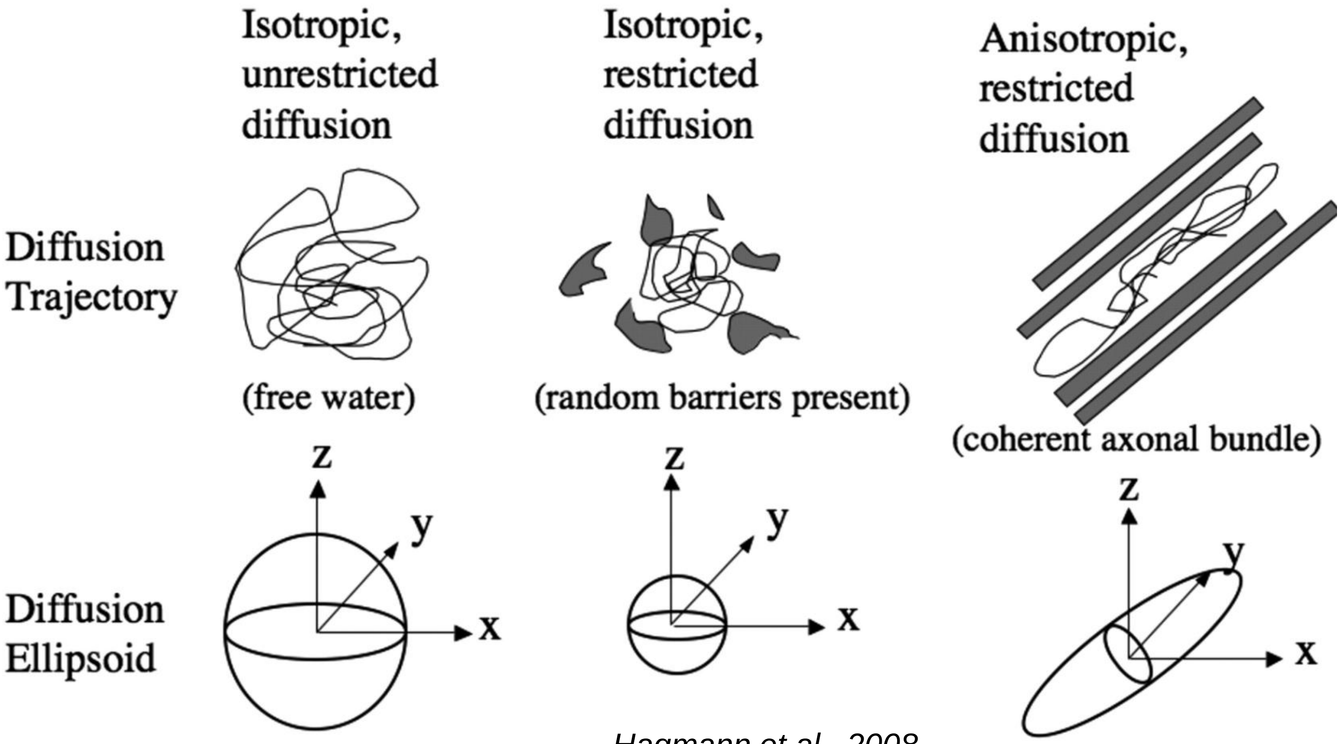


Hemodynamic changes associated with brain activity

Water Diffusion Reveals White Matter Organization



Water Diffusion



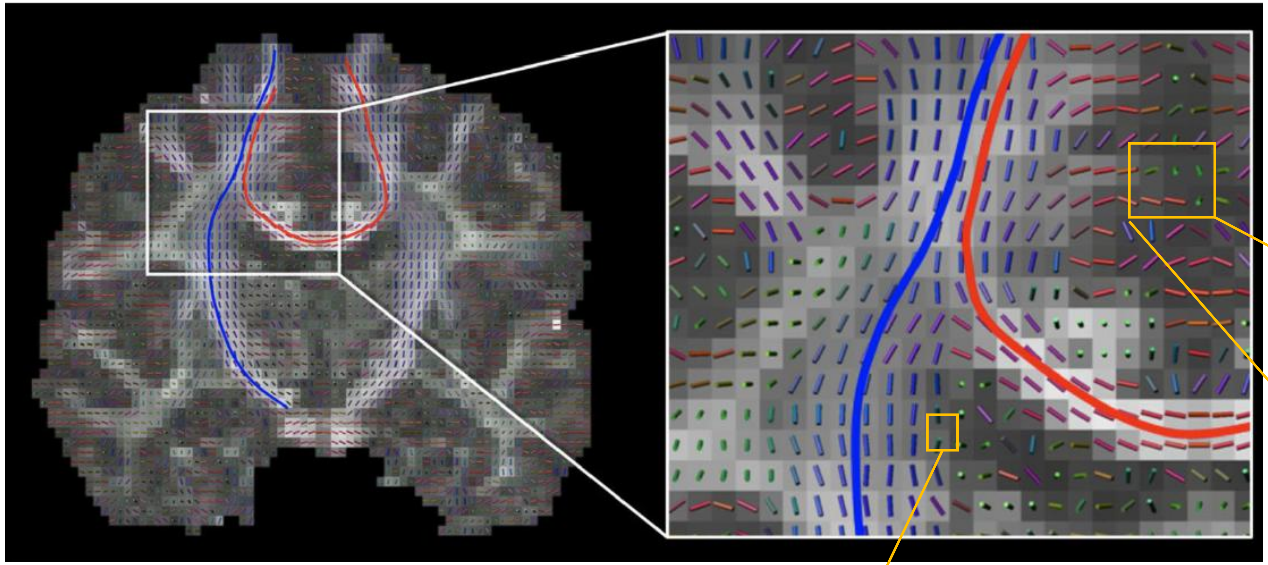
Hagmann et al., 2008

The direction of water diffusion provides indirect information about white matter organization.

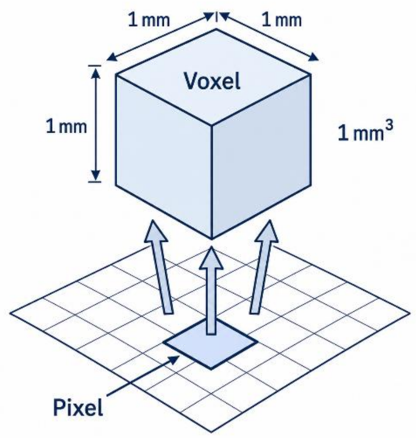
Local Diffusion Models Estimate Fiber Orientations

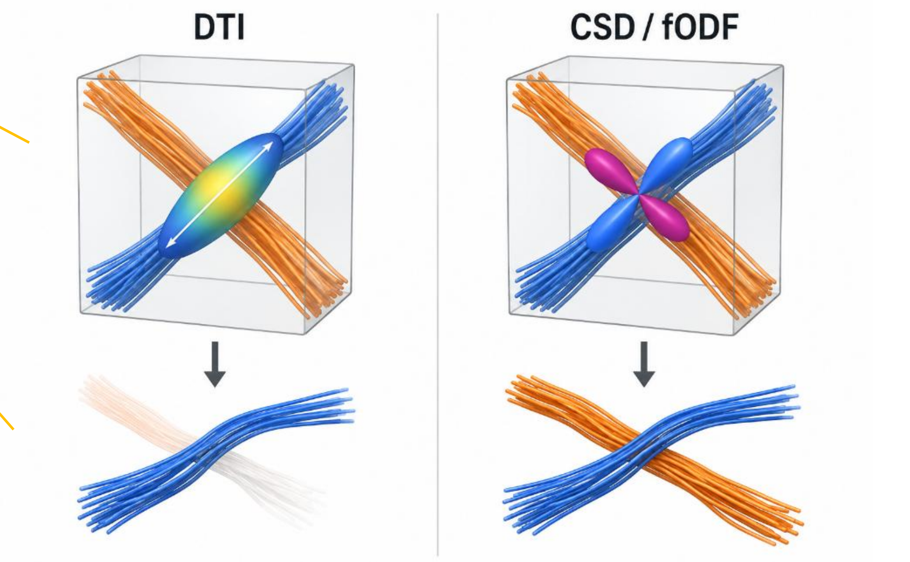
How difussion information is represented in an MRI image?

Diffusion Models



Jeurissen et al.,





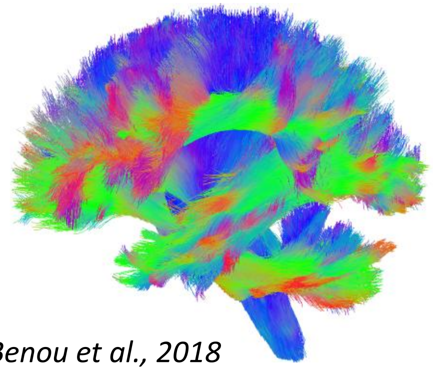
Estimate one main direction

Estimate several directions

Reconstructing White Matter Pathways

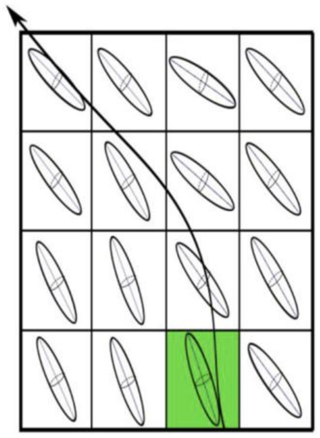
Local diffusion models estimate one or more fiber directions in each voxel, and then tractography connects them across neighboring voxels to reconstruct white matter pathways.

Tractography



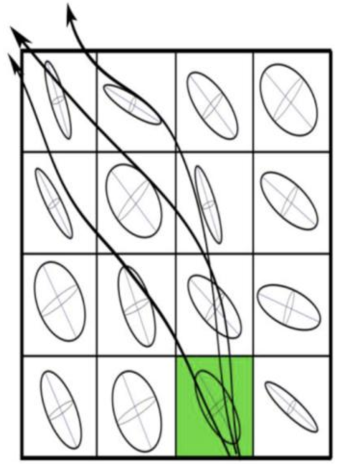
Benou et al., 2018

Deterministic



Follows the single most probable fiber orientation step-by-step.

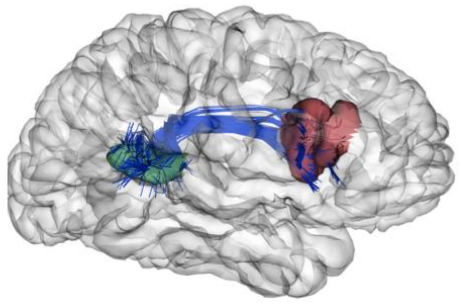
Probabilistic



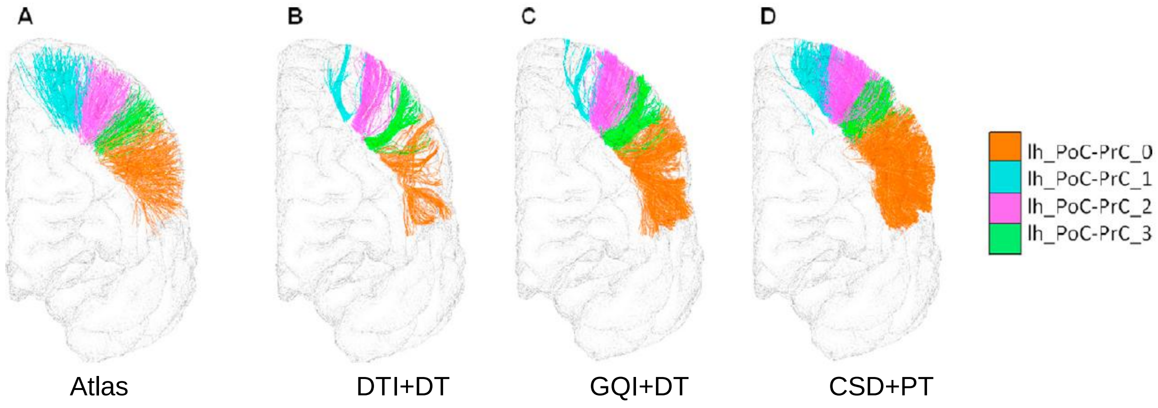
Generates multiple streamlines per seed point by sampling a distribution of possible directions

Brain Bundles

Seitz-Holland 2016 [11]



López 2020 [8]

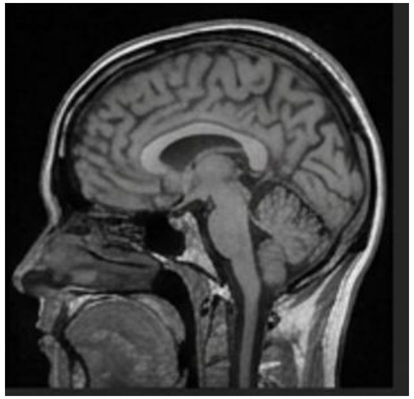


Guevara et al., 2018

Cortical Parcellation

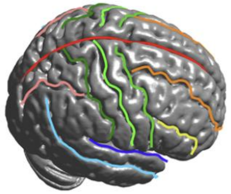
Methods that divide the cortex into regions by grouping points with similar characteristics

Different Criteria for Cortical Parcellation

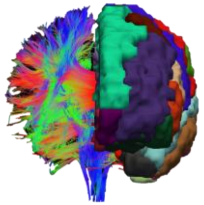


sMRI
(to reconstruct the cortex)

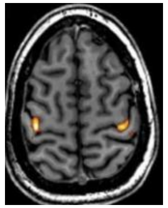
Anatomical



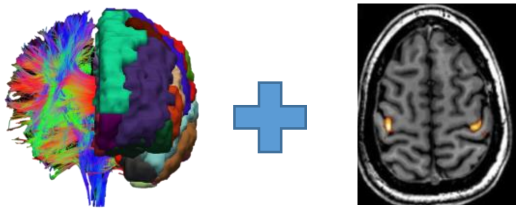
Connectivity-Based



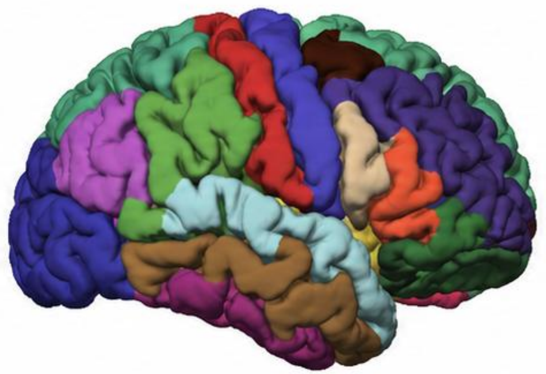
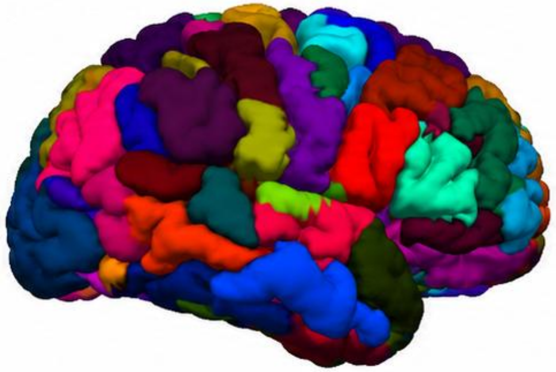
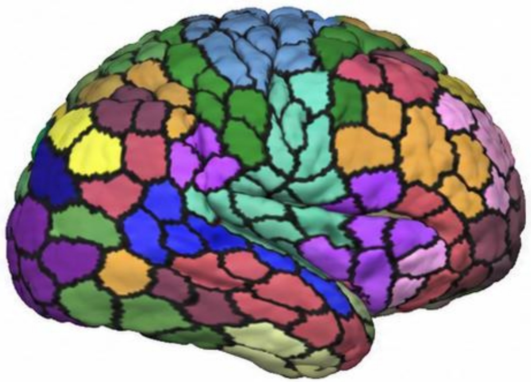
Functional



Multimodal



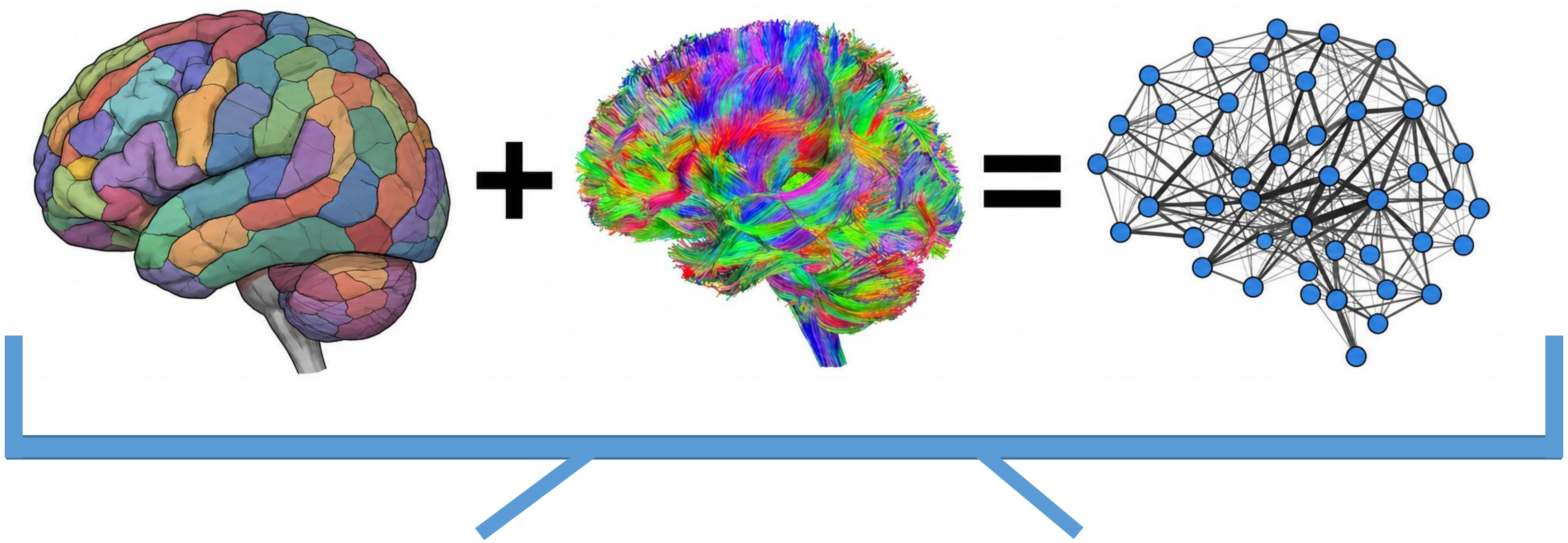
Brain Atlases: Different Ways to Parcellate the Cortex

Desikan-Killiany	AAL	Brainnetome	Shaefer
			
Anatomical surface-based atlas defined by cortical gyri and sulci, comprising 68 regions	Anatomical atlas that divides the brain into 116 regions (90 without the cerebellum regions)	Connectivity-based atlas with 210 cortical and 36 subcortical regions	Cortical atlas derived from resting-state functional connectivity and available at multiple resolutions

No single atlas is suitable for every study

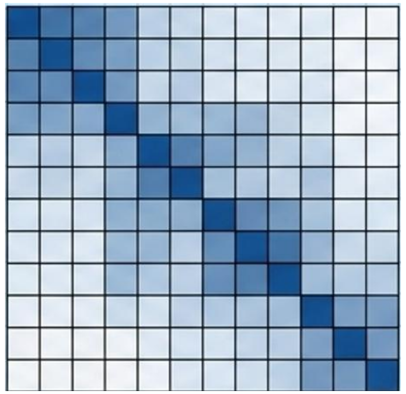
Constructing the structural Connectome

Cortical Parcellation　　Tractography　　Structural Connectome



Connectivity Matrix

$$c_{nm} = \frac{2F_{nm}}{V_n + V_m}$$

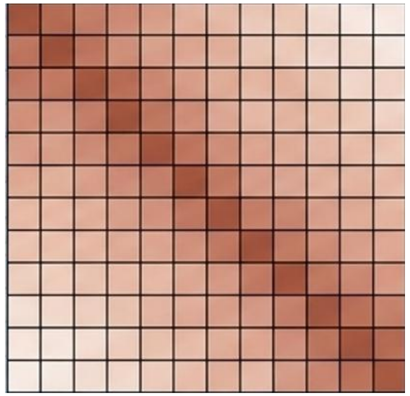


c_{nm} : connection weight between regions *n* and *m*

F_{nm} : number of streamlines connecting regions *n* and *m*

V_n, V_m: volumes of the two regions

Time-Delay Matrix



$$\tau_{nm} = \frac{L_{nm}}{v}$$

τ_{nm}: time required for a signal to travel between regions *n* and *m*

L_{nm} : mean streamline length between the regions *n* and *m*

v : conduction velocity

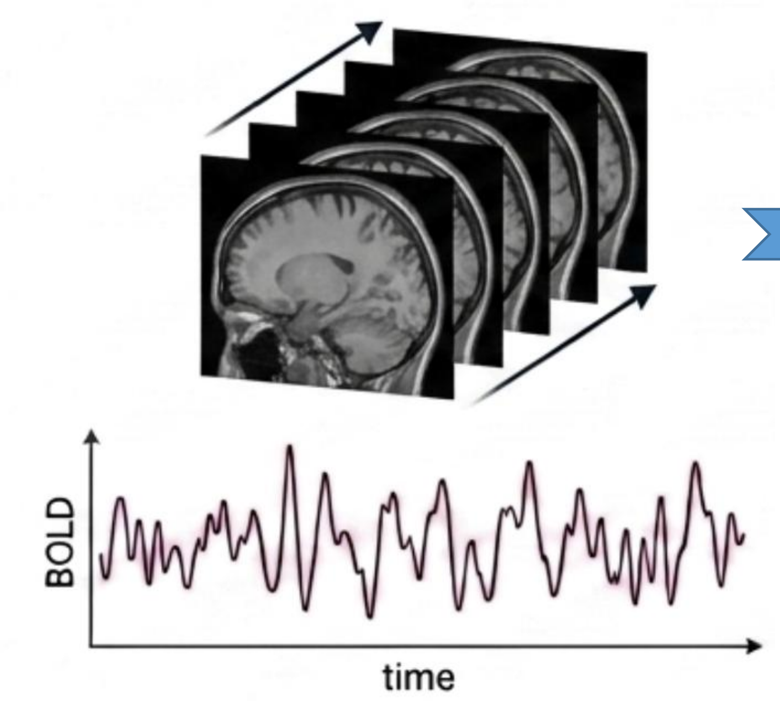
Constructing the Functional Connectome

To study how the activity of different regions is related

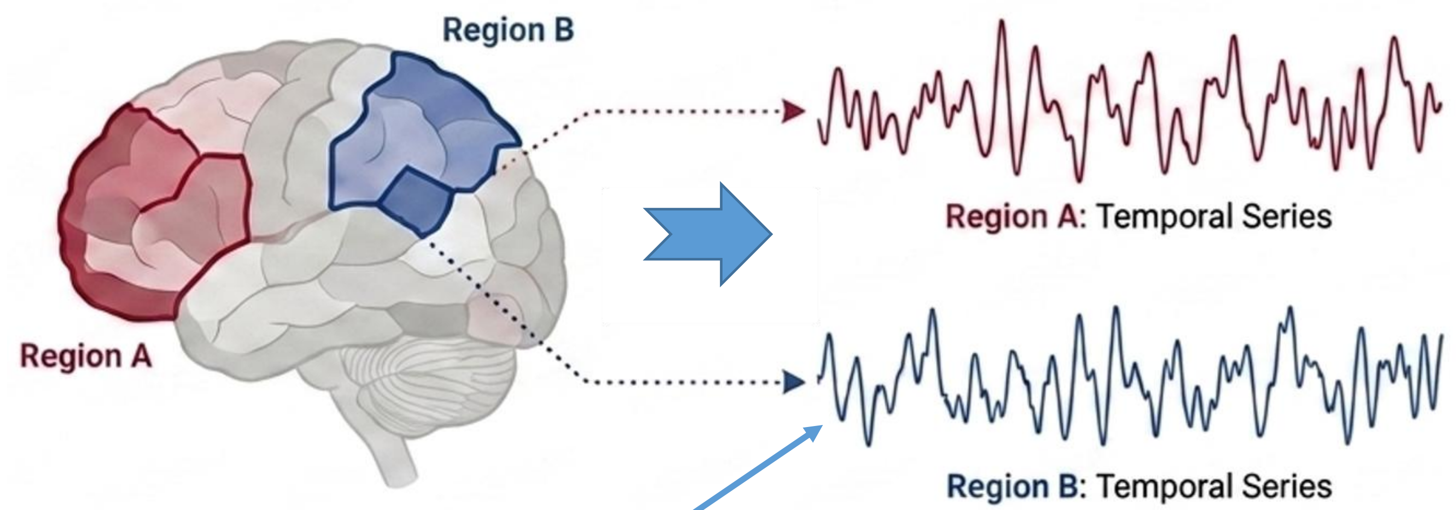
Acquisition

Brain Parcellation

Temporal Series Extraction

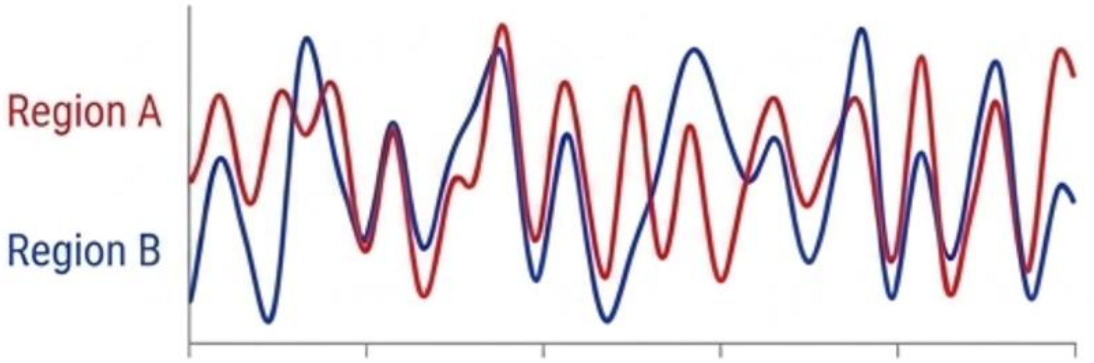


Changes in de blood oxygenation
Related to brain activity



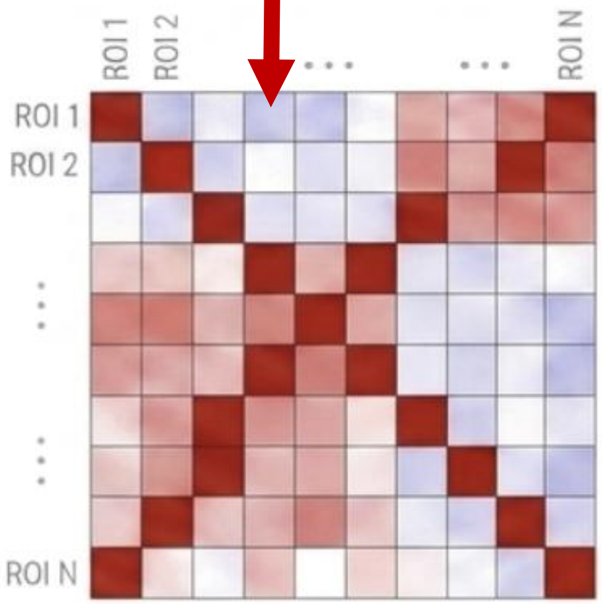
BOLD signals from the voxels are averaged for each region

Constructing the Functional Connectome



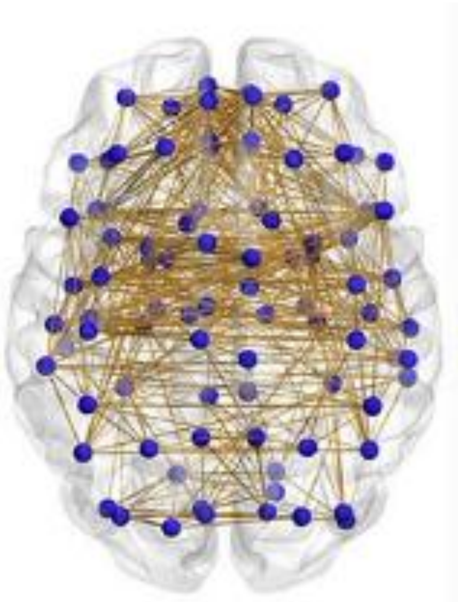
Similarity between two signals over time

Pearson correlation (r)



1 (close to strong correlation)
0 (close to weak correlation)
-1(Opposite changes)

Functional connectivity matrix
(strength of the functional connectivity)



Functional Connectome
(Graph Representation)

Software Tools for Connectome Reconstruction

Tractography and structural connectivity

- **MRtrix3:** Advanced tractography and connectome construction
- **FSL:** Diffusion preprocessing and probabilistic tractography
- **DSI Studio:** GUI-based reconstruction and fiber tracking
- **DIPY:** Custom dMRI and tractography pipelines in Python
- **TractSeg:** Automatic white-matter bundle segmentation

Parcellation and functional analysis

- **FreeSurfer :** Anatomical cortical parcellation from T1 MRI
- **Nilearn:** Brain atlases, functional parcellation and connectivity

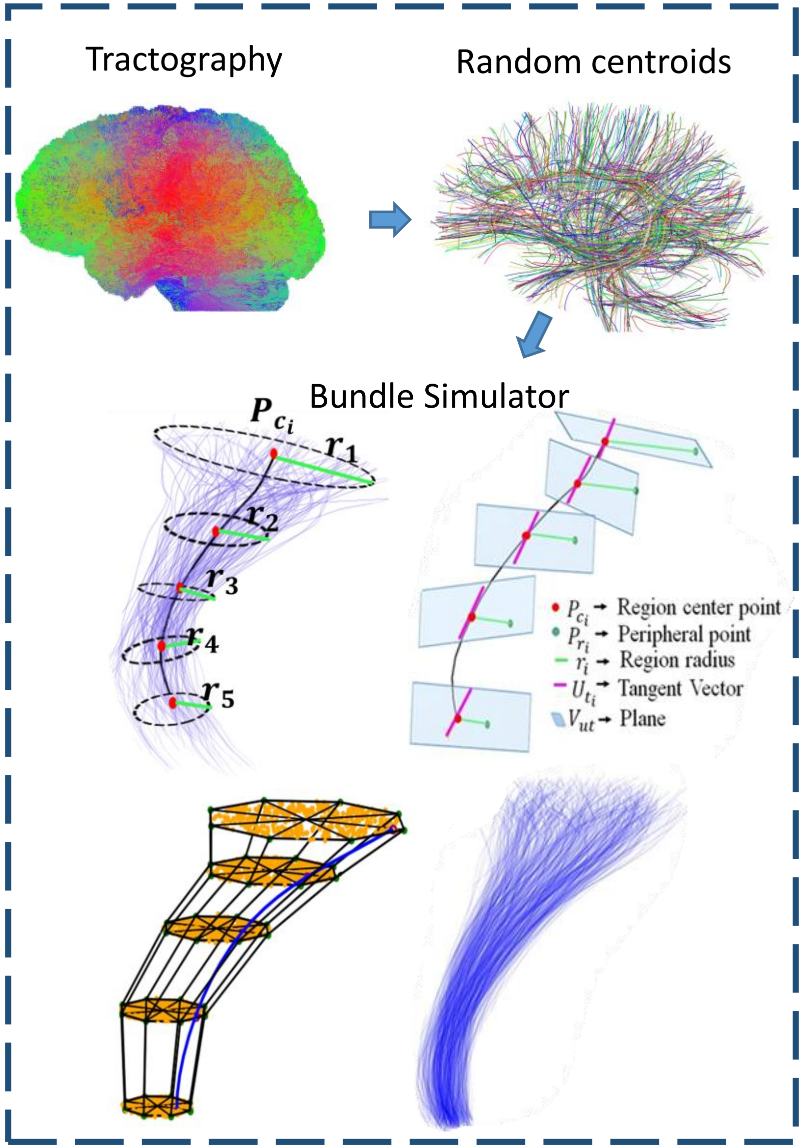
Freesurfer

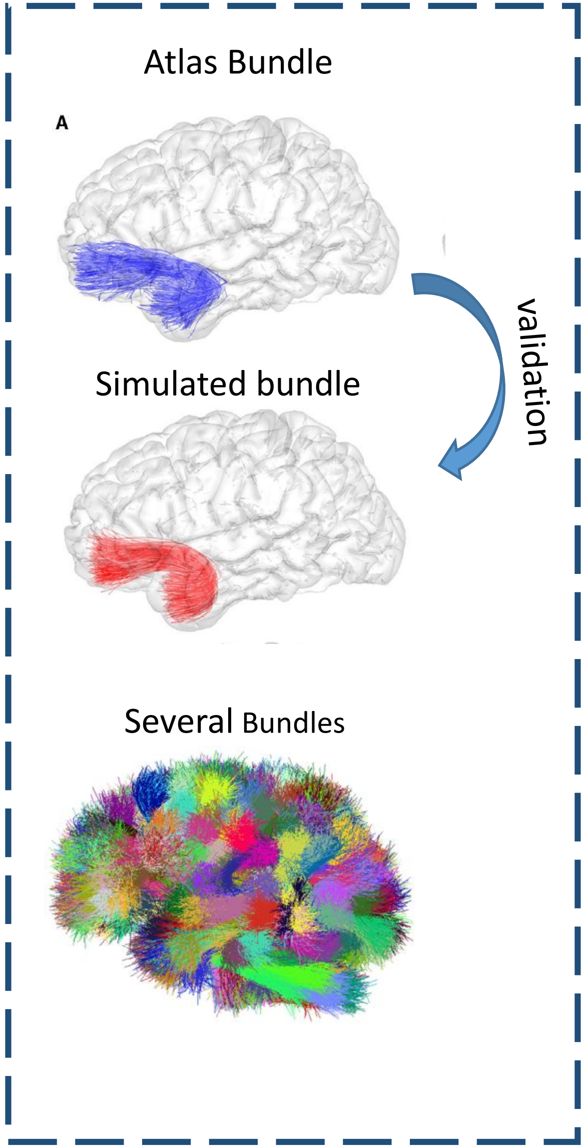
https://www.youtube.com/watch?v=6wxJ1up-E7E&t=140s

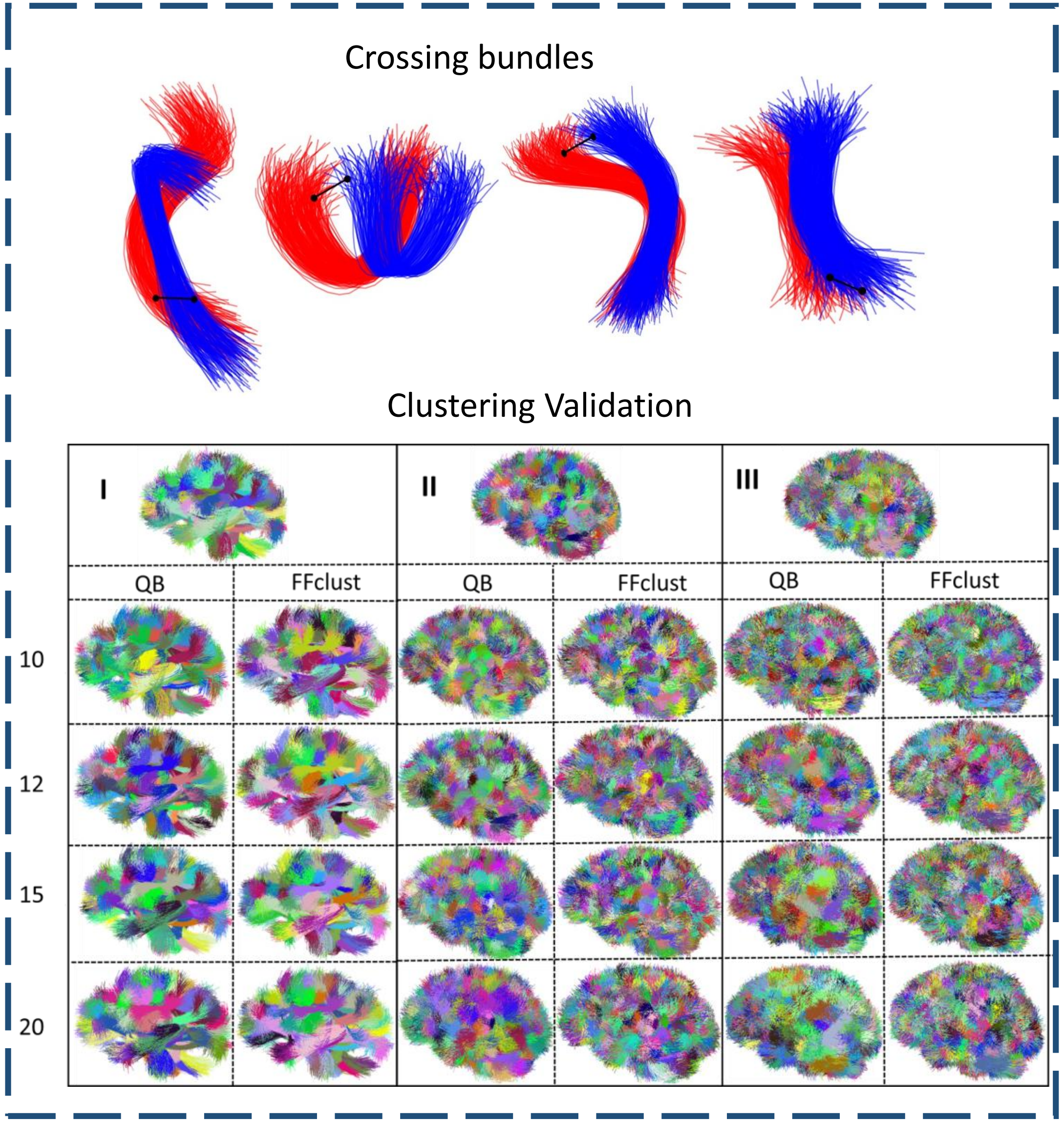
MRtrix3

https://www.youtube.com/watch?v=DK2ZqJd9yZE&list=PLIQIswOrUH68Zi9SVDAdcUExpq2i6A2eD&index=7

PhyberSIM: A tool for the generation of ground truth to evaluate brain fiber clustering algorithms



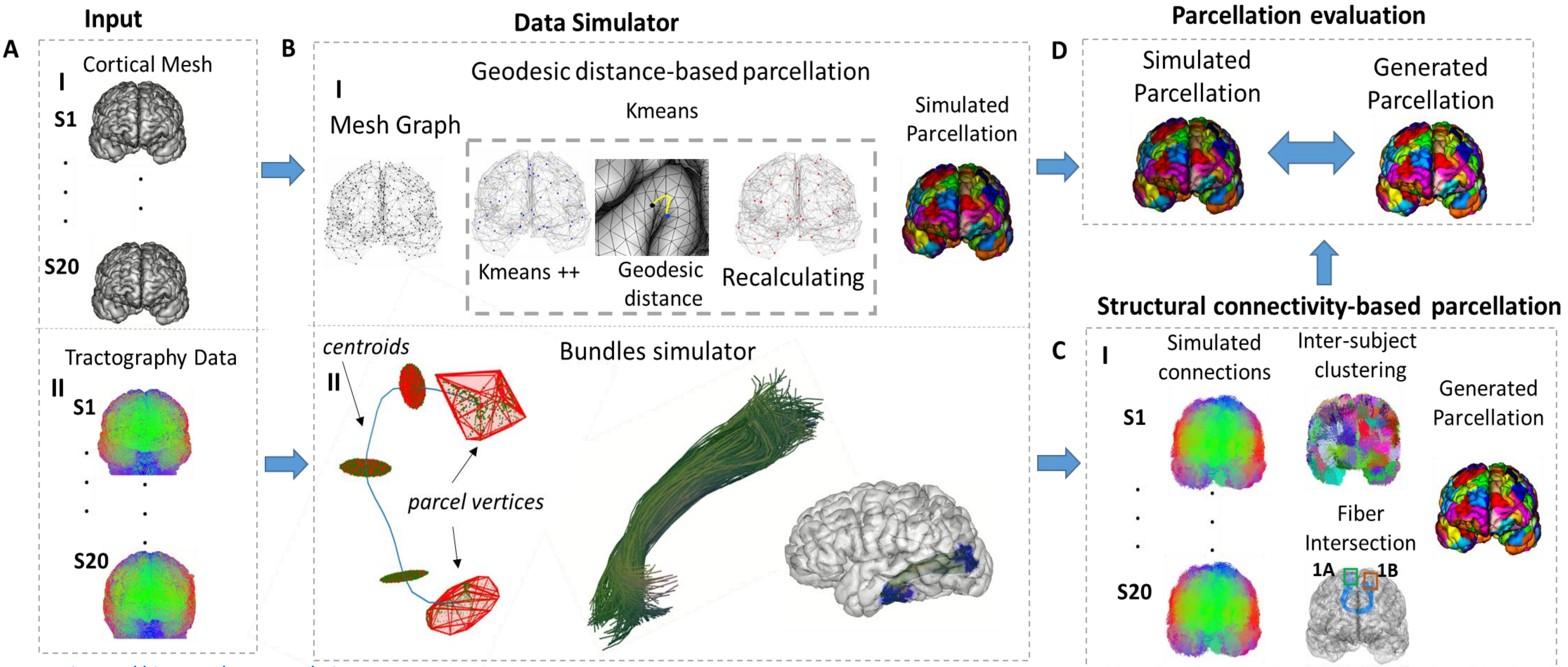




https://doi.org/10.1117/12.2669811
https://doi.org/10.3389/fnins.2024.1396518
https://github.com/elidapoo/Brain_bundle_simulator

Poo et al.,2025

A simulator for the validation of tractography-based cortical surface parcellations



https://doi.org/10.1016/j.ibneur.2023.08.1611

https://doi.org/10.1016/j.compbiomed.2025.110891

https://github.com/elidapoo/Brain_bundle_simulator_for_parcellation

Poo et all.,2025

GRACIAS!!!